





Quaid-e-Awam University of Engineering, Science and Technology ,Nawabshah



Synthesis and Characterization of Chemically Modified Polyacrylonitrile (PAN) Fiber for Arsenic Removal from Synthetic Water

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Content table



- Introduction
- Literature review
- Problem statement
- Research gap
- Scope of work
- Aim & Objective
- Material and Methodology
- Results and Discussion
- Conclusion
- References

Introduction



Water is a vital natural resource that needs to be protected from pollutants[1].

As the population grows and climate change affects the environment more, the supply of fresh and clean drinking water is decreasing.

Over **785 million** people in the world lack access to clean drinking water.

About **800 children** die from diarrhea every day in the world due to poor water quality[2].

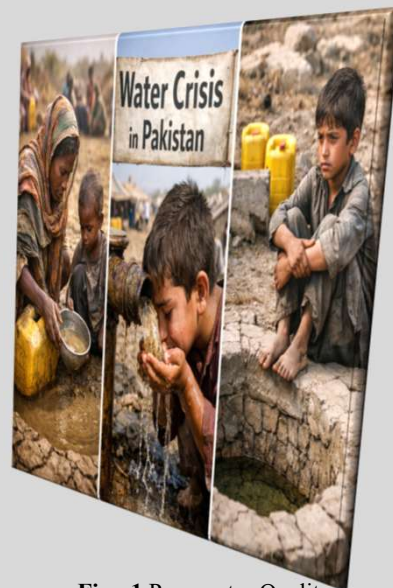


Fig : 1 Poor water Quality

Water Pollution in Pakistan

Pakistan is facing worsening water scarcity and contamination, making clean drinking water a critical health concern [3].

According to the recent global environment performance index (EPI) ranking. Pakistan is one of the country that faces serious water pollution, water scarcity, poor air quality, global warming, and climate change [2].

Water quality is also affected due to various human activities such as industrial and agricultural expansion. As Shown in Fig:2 (a , b)

Pollutants such as chemicals, microbial organisms, industrial wastes, and toxic substances contribute to unsafe drinking water in the country [3].



Fig : 2 (a)

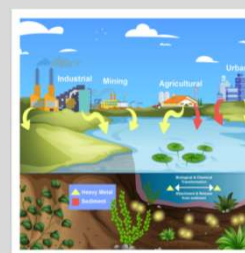


Fig : 2 (b)

Major contaminants in drinking water

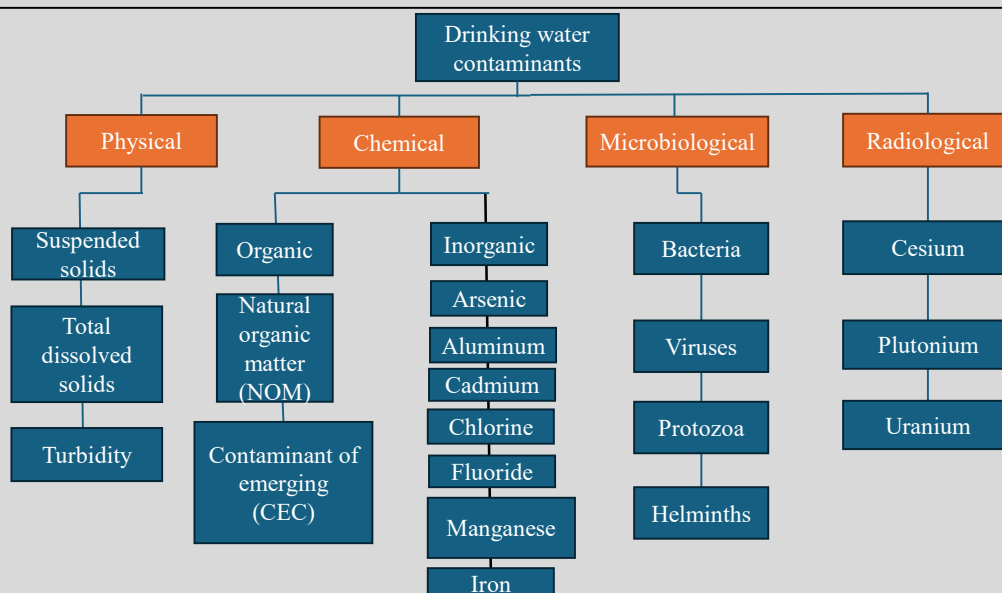


Fig. 3. Major contaminants in drinking water considered [6]

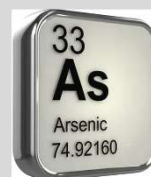
Arsenic

Arsenic pollution is mainly due to natural resources.

Arsenic is a poisonous element that can cause cancer. It's the 20th most common element on Earth and comes in two main forms: Arsenite (As^{3+}) and arsenate (As^{5+}) [1,4,5].

In order to reduce arsenic exposure, the WHO has lowered the arsenic maximum contaminant level in the water to **10 ppb**[6].

The set value suggested by the Environmental Protection Agency of Pakistan (EPA) that is $50\mu\text{g/L}$.



Problems of arsenic in Pakistan

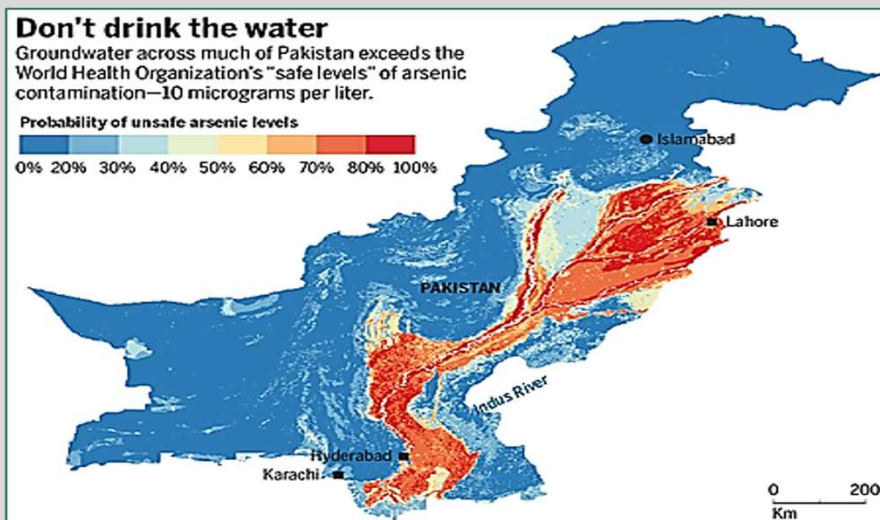


Fig : 4 map shows arsenic level in Pakistan in which 26% in Sindh 36% Punjab ,1.2% and 22 in KPK [31].

Problems of arsenic in Sindh

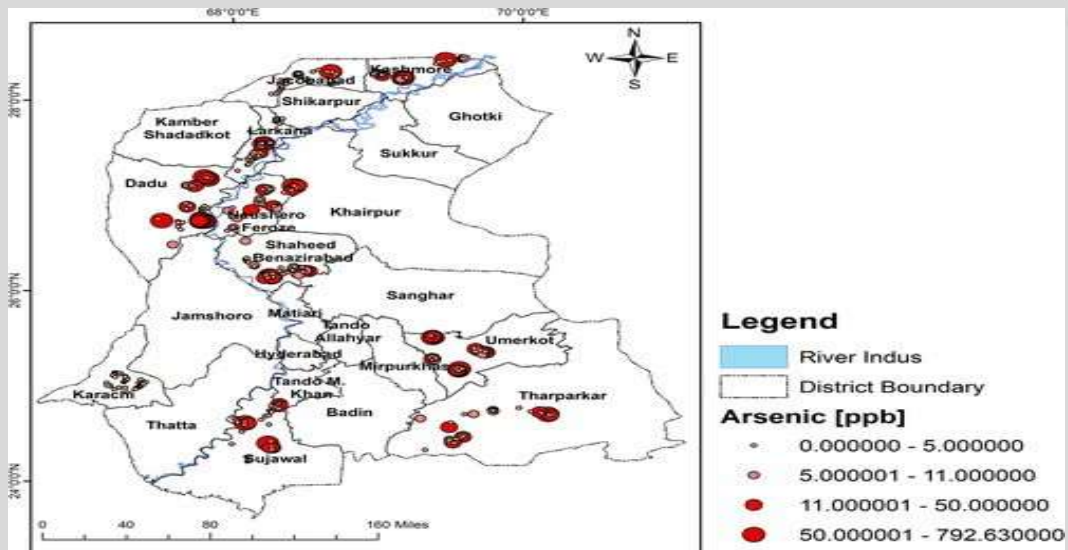


Figure:5 arsenic problems in Sindh province [6]

Arsenic Removal Technologies

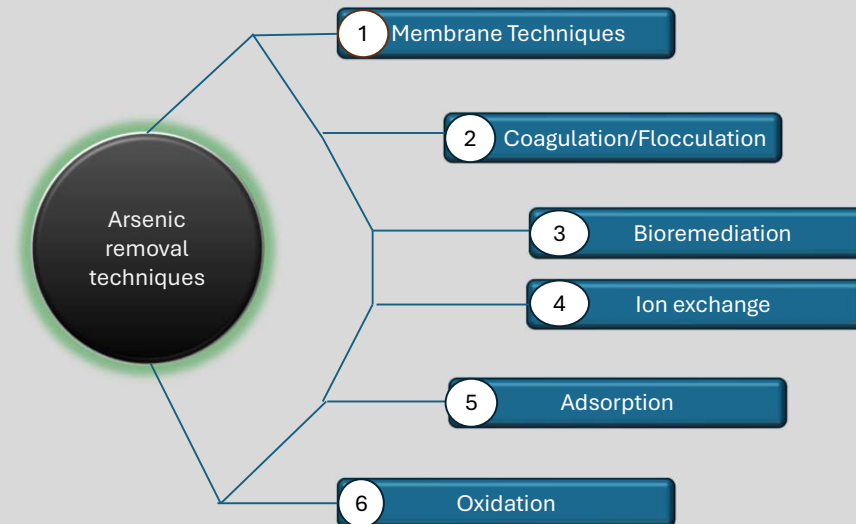


Fig : 6 Different Technologies for arsenic Removal [9]

Adsorption Technology



Adsorption is a simple and effective method used in wastewater treatment. It helps clean water by trapping both organic and inorganic pollutants on the surface of special materials [7].

Adsorption is a low-cost and fast method for water treatment. It can remove even very small amounts of pollutants effectively if the adsorbent material has high capacity and strong attraction to contaminants [8].

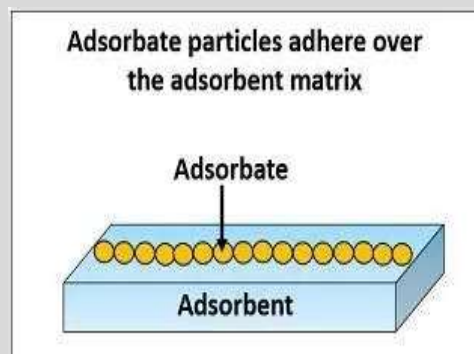


Fig : 7 Adsorbent mechanism

Polyacrylonitrile (PAN) Fiber and its Characteristics



(PAN) fiber widely used in textiles and carbon fiber production, can be chemically modified by transforming nitrile groups into functional groups (amines, carboxyl, amidoxime etc.). PAN fiber shown in Fig:8. These modifications enhance adsorption capacity for removal of heavy metals such as (e.g. As, Cd, Pb, Cr, and etc.) from water.

PAN has properties involving high boiling point (317°C), large surface area and its limited solubility of PAN in some solvents [3].



Fig: 8 PAN Fiber in Raw or Un-functionalized Form.

Literature review			
S.No	Author	Year	Research summary
1	Rahman, md mahbubur, et al.	2025	Reviewed the role of bio-based nanomaterials in groundwater arsenic remediation, focusing on mechanisms, challenges, and future perspectives. The study highlights eco-friendly, cost-effective nanostructures derived from biological sources as sustainable alternatives to conventional adsorbents [10].
2	Chen, yongqiang et al.	2025	Reviewed recent advancements in adsorption and detection of wastewater pollutants using covalently functionalized polyacrylonitrile fibers (PANF). PANF, widely used in textiles and carbon fiber production, can be chemically modified by transforming nitrile groups into functional groups (amines, carboxyl, amidoxime, quaternary ammonium salts). These modifications enhance adsorption capacity for heavy metals such as Hg^{2+} , Fe^{3+} , Cu^{2+} , Ni^{2+} , Pb^{2+} , Cd^{2+} and Cr^{3+} , as well as organic pollutants and dyes [11].
3	Alam, E.	2025	Provided a comprehensive overview of metal–organic frameworks (mofs) for arsenic remediation, highlighting their structural tunability, large surface area, and adjustable pore sizes as key advantages for water treatment. Mofs are crystalline porous materials composed of metal ions/clusters coordinated to organic ligands, enabling selective adsorption of contaminants such as arsenic [12].

Literature review			
S.No	Author	Year	Research summary
4	Satyam & patra	2024	Presented a comprehensive review of adsorption-based wastewater remediation, highlighting historical developments, recent innovations, and persistent challenges. The study emphasizes the role of nanotechnology, advanced fabrication methods, and AI/ML integration in designing next-generation adsorbents for sustainable water purification[13].
5	Kaveh & jazani et al.	2024	Investigated the design, preparation, and characterization of modified pan-based carbon fibers using an anodic electrochemical oxidation process. The study focused on the role of electrolytes in altering fiber morphology and surface properties to improve adhesion in composite applications[14].
6	Shokri & naghashzargar et al	2024	Developed adsorptive electro spun nanofibrous membranes (aenms) based on polyacrylonitrile (PAN) incorporated with cloister 20A (C20) organoclay for arsenic (as(v)) removal from contaminated water. The study investigated structural properties, adsorption performance, and regeneration ability of the fabricated membranes[15].

Literature review			
s.no	Author name	Year	Research summary
6	Memon & Unar et al.,	2023	investigated arsenic adsorption using a packed bed column designed with chemically modified polyacrylonitrile (PAN) fiber. The adsorbent was prepared by coating PAN fiber with chitosan/iron oxide, and its performance was evaluated through simulation and modeling studies using Aspen Adsorber V11. Both small-scale (2 cm diameter, bed depths 1–3 cm, flow rates 1–3 mL/min) and large-scale (8 cm diameter, bed depth 8 cm, flow rates 512–1536 mL/min) columns were tested [16].
7	Dilpazeer & Munir et al.,	2023	presented a comprehensive review of arsenic contamination in groundwater, its sources, health impacts, and the latest removal technologies. Arsenic is recognized as a Class-1 carcinogen by WHO, with global exposure affecting more than 300 million people, particularly in Asia[17].

S.No	Author	Year	Research summary
8	Cheng & Zhang et al	2022	Developed phosphonic acid functionalized polyacrylonitrile (PPAN) hollow fiber membranes via a mannich reaction to enhance heavy metal adsorption. The study focused on Hg^{2+} removal from aqueous solutions, evaluating adsorption capacity, kinetics, isotherms, and reusability [18].
9	Gupta & Sharma et al.	2022	Reviewed the role of polyacrylonitrile (PAN) as a versatile and cost-effective polymer for fabricating adsorbents used in wastewater treatment. The paper highlights PAN's structural properties, functionalization methods, and applications in removing heavy metals and dyes from contaminated water. Where in the methodology. Forms of PAN adsorbents: ion exchange resins, chelating resins, fibers, membranes, and hydrogels. Synthesis techniques: suspension polymerization, emulsion polymerization, redox polymerization, grafting, and electrospinning. Functionalization: PAN modified with ligands such as amidoxime, polyethylene imine, acrylamide, thiazoline, amino acids, and thiourea to enhance selectivity and adsorption capacity[19].
10	Zheng, Liwei.	2021	Investigated reusable electro spun carbon nanofiber composites for the selective removal of inorganic arsenic species ($As(III)$ and $As(V)$) from water. The study focused on developing high-performance, regenerable adsorbents using nanofiber technology. Where methods are used, material preparation: electrospinning of carbon nanofibers combined with functional additives to enhance arsenic adsorption. Characterization techniques: SEM (morphology), FTIR (functional groups), XRD (structural analysis), BET (surface area), and contact angle (hydrophilicity). Adsorption studies: batch and dynamic experiments for both $As(III)$ and $As(V)$ [20].

Problem statement



Arsenic contamination in drinking water is a serious health issue worldwide, especially in Sindh Pakistan. Groundwater often contains arsenic levels above WHO safety limits. Due to presence of arsenic at high level can negatively impact on our health such as cancer, skin disease, and long-term organ damage. Therefore, Adsorption technique is recommended to produce safe drinking water.

- Now current challenges are in conventional water treatment methods that it often failed to remove arsenic effectively, at low concentration.
- So, there is the demand for affordable, efficient and sustainable adsorbent material which remove arsenic at low concentration.
- Therefore, the novelty of this work is modification of PAN fiber adsorbent material which can increase removal efficiency of arsenic at low concentration there by decreasing process cost.



Research Gap



- PAN fiber are widely used in the various industries such as textile, carbon fiber production, filtration membrane and biomedical.
- However, Polyacrylonitrile (PAN) fibers are promising in their applications , but chemically modified versions for arsenic removal are not yet fully explored.
- Therefore, modification and synthesizes of Polyacrylonitrile Fiber can increase the arsenic removal efficiency in that way decrease the process cost have been investigated to overcome those drawbacks, and display good arsenic adsorption capacity.

Scope of Work



- This study focused on the polyacrylonitrile (PAN) fiber, which was chemically modified by introducing functional groups.
- The modified fiber was analyzed through Fourier Transform Infrared Spectroscopy (FTIR) and Scanning Electron Microscopy (SEM) for characterization.
- The modified PAN was used in column experiments for arsenic removal.
- Atomic Absorption Spectroscopy was conducted to analyze the adsorption capacity of the modified PAN fiber.

Aim of work : To remove arsenic from synthetic water using chemically modified PAN fiber adsorbent

Objectives



01

To develop chemically modified polyacrylonitrile (PAN) fiber

02

To characterize the chemically modified PAN fiber

03

To analyze and evaluate the efficiency of Chemically Modified PAN fiber for Arsenic (A^{+5}) removal through column study.

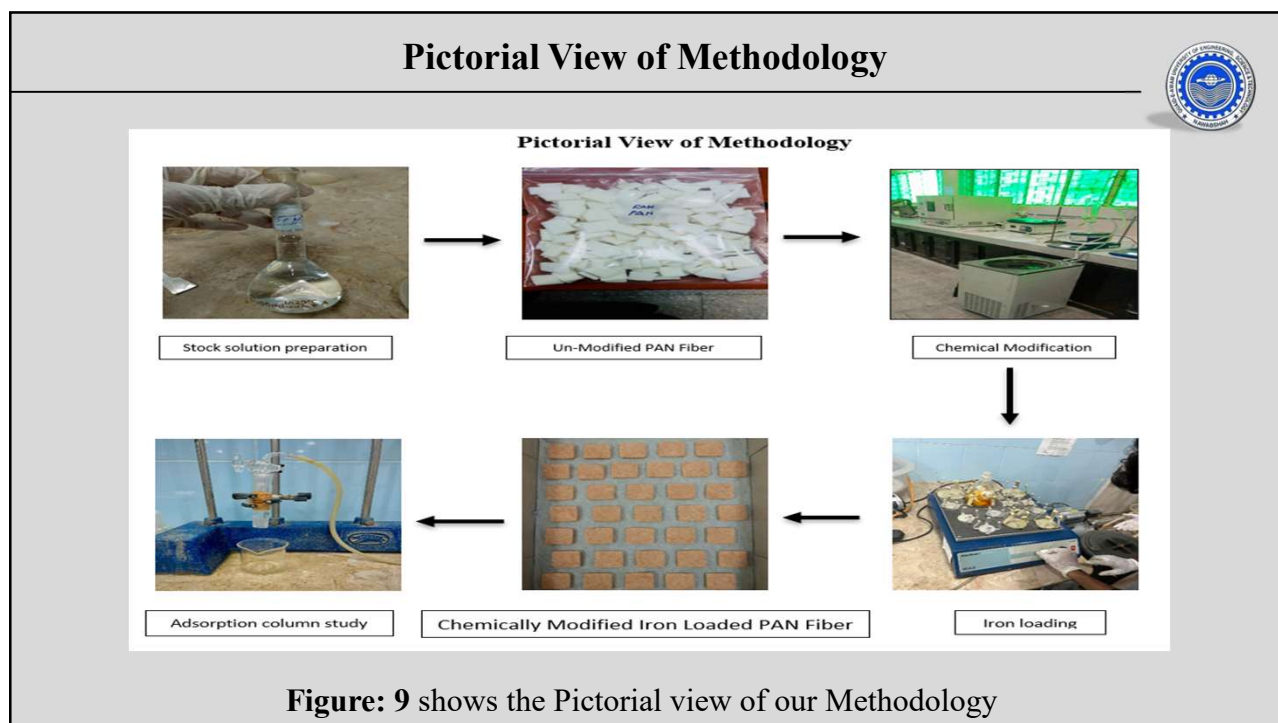
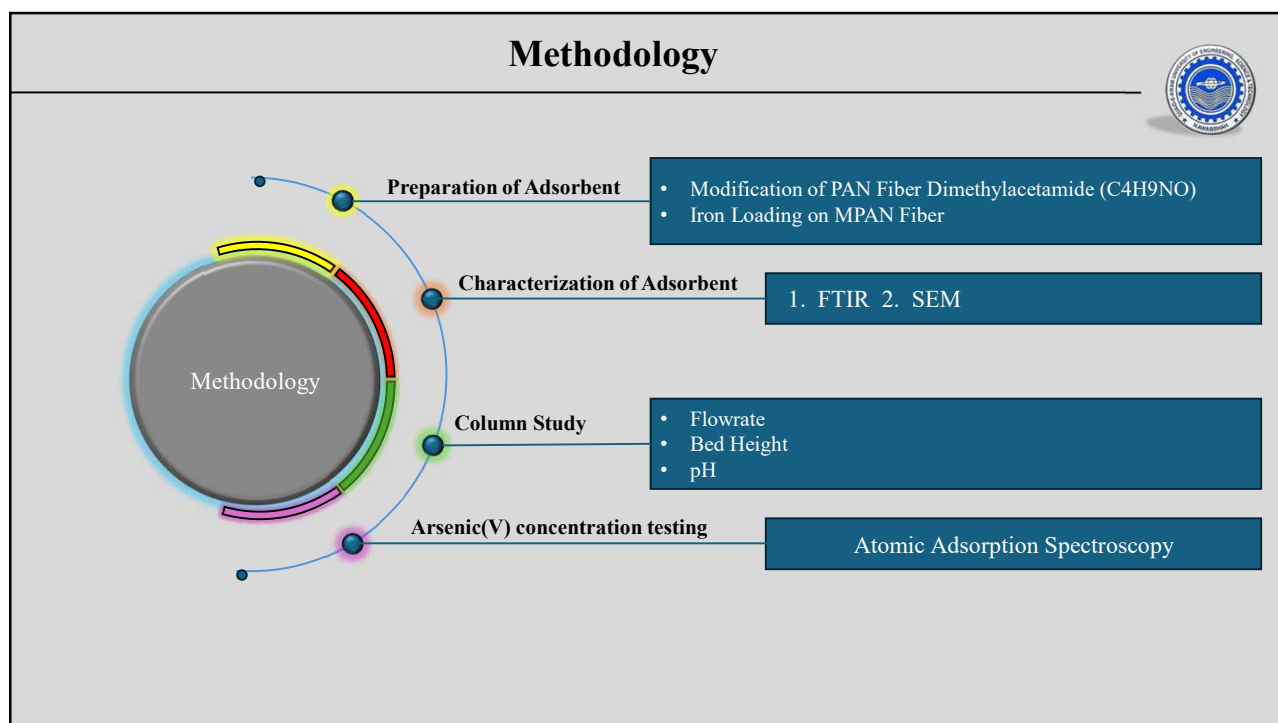


Figure: 9 shows the Pictorial view of our Methodology

Material & Methodology



OBJECTIVE 1

Adsorbent Preparation

Material & Methodology



➤ Modification of PAN Fiber

- PAN fiber was purchased from Heading Filter Material China.
- PAN fiber modified by using Dimethylacetamide (C_4H_9NO). 10 g of PAN fiber was taken and put in 500mL round bottom flask then added 250mL of DMAC and 5 g of Aluminum chloride hexahydrate.
- The flask was equipped with heating mantle and a reflux condenser mounted on magnetic stirrer plate as shown in Fig. 10
- PAN fiber was heated for 8 hours at 90 °C and stirred at 340 rpm. Temperature and mixing were continuously monitored and controlled.
- After 8 hrs heating, pH was checked and found 13, MPAN fiber samples were washed with distilled water repeatedly and the pH after washing was 10. Fig : 10



Fig : 10 chemical Modification

Material & Methodology



➤ Iron loading on Modified PAN fiber

- MPAN fiber was loaded with ferric chloride hexahydrate ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$).

1. Preparation of Iron Solution

- 13.5 g $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ was dissolved in 250 mL distilled water to make 0.2 M FeCl_3 solution. And then it was Stirred until fully dissolved. Shown in Fig:11 (a,b).

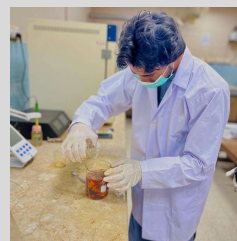


Fig:11 (a) Iron solution preparation



Fig:11 (b) weighting Ferric Chloride

Material & Methodology



2. Impregnation (Iron Loading)

- Modified PAN Fiber was immersed in the 250 ml FeCl_3 solution. Fig:12 (a)
- Modified PAN Fiber was Stirred gently at room temperature for 3 hours. Fig:12 (b)
- After stirring 3 hours the then Slowly dropwise 0.5 M NaOH solution (~200 mL) was added to the Modified PAN-iron mixture while stirring. For pH maintaince.
- pH was maintained up 8-10 using pH meter , and then 1.5 hours stirred continuously. Fig:12 (b)



Fig:12(a) Dumping M-PAN in solution



Fig:12 (b) Stirring gently



Fig:12(c) Maintaining pH by pH meter

Material & Methodology



3. Aging

The mixture was Leaved undisturbed for 12 hours at room temperature. Over at night time For to improve coating stability and crystallinity.

4. Washing

Iron coated MPAN Fiber was washed 5–6 times with ~150 mL distilled water each wash. This process Continued until wash water was reached to neutral pH (~7). As shown in Fig:13 (a)

5. Drying

Iron-loaded Modified PAN Fiber was dried in the Oven for 6 hours. Shown in Fig: 13 (b) and then it was leaved over night at room temperature and then it was fully prepared to use as adsorbent. Fig : 13 (c)



Fig :13 (a) Washing of iron loaded M-PAN fiber



Fig :13(b) Dry in Ovan at 80C



Fig: 13 (c) Fully Prepared Adsorbent for use

Material & Methodology



OBJECTIVE 2

Characterization of Adsorbent

(1) *Fourier Transform Infrared Spectroscopy*

(2) *Scanning Electron Microscopy (SEM)*

Material & Methodology



(1) Fourier Transform Infrared Spectroscopy

Fourier Transform Infrared Spectroscopy (FTIR) is a technique used to identify chemical bonds and functional groups in a material. Its shown in Fig : 14

How it works.

- Infrared light is passed through a sample, and certain wavelengths are absorbed depending on the chemical bonds present.
- The instrument produces a spectrum showing absorption peaks, each corresponding to specific functional groups [3].



Fig: 14 (FTIR) Machine

Material & Methodology



(2) Scanning Electron Microscopy (SEM)

Scanning Electron Microscopy (SEM):

- Uses a focused beam of electrons.
- Produces high-resolution images of surfaces.
- Shows texture, shape, and composition.
- Magnification up to nanometer scale.
- Widely used in materials science, biology, and engineering. Machine shown in Fig: 15



Fig:15 (SEM) Machine

Material & Methodology



OBJECTIVE 3

Fixed Bed Adsorption Column study

Material & Methodology



➤ Preparation of stock solution As (V) 50ppb

- As(V) stock solution was prepared by dissolving 0.24029g potassium arsenate monobasic (H_2AsKO_4) in 100ml distilled water to make 1000ppm.
- From 1000ppm stock solution took 10ml in 100ml distilled water to make 100ppm. For 1ppm took 1ml of 100ppm stock solution in 100ml distilled. For 50ppb took 5ml of 1ppm stock solution in 100ml distilled water. Overall preparation of solution process are shown in fig:16 (a,b,c)



Fig:16 (a & b) making solution for 50 ppb

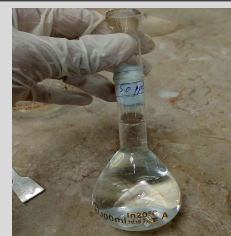


Fig:16 (c) Prepared solution of 50 ppb

Material & Methodology



➤ Lab-scale Experiments for removing Arsenic Fixed bed column.

- Continuous fixed bed column was setup to conducted different experiments. To investigated the column break through and study the effects of process parameters in fixed bed column such as Inlet flow rate, adsorbent bed height and pH. Fig:17(c).



Fig:17 (a & b) Changing different Parameters (flowrate, Bed Height & pH.

- Conducted of various experiments to analyze the effects of following parameters on Arsenic removal efficiency of synthetic Arsenate water. Fig: 17 (a & b)

- Effect of Flowrate
- Effect of bed height
- Effect of pH



Figure : 15 Adsorption Column Study

Results and Discussion



➤ Fourier Transform Infrared Spectroscopy

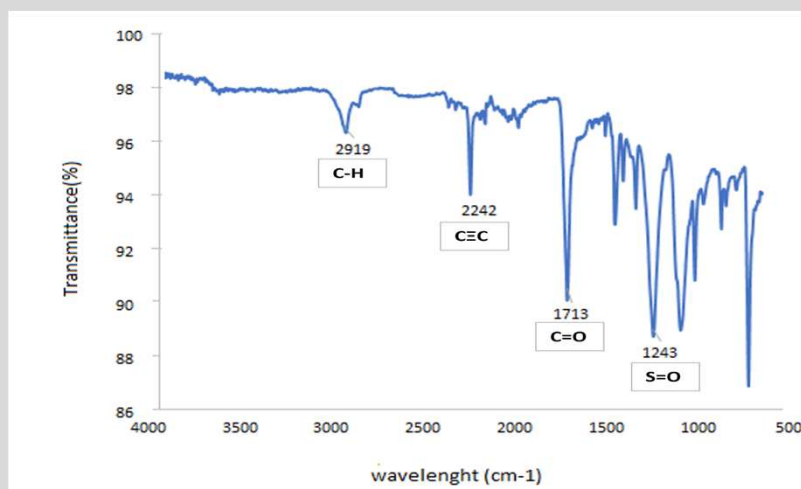


Figure : 18 Un Modified PAN Fiber

Results and Discussion

➤ Fourier Transform Infrared Spectroscopy

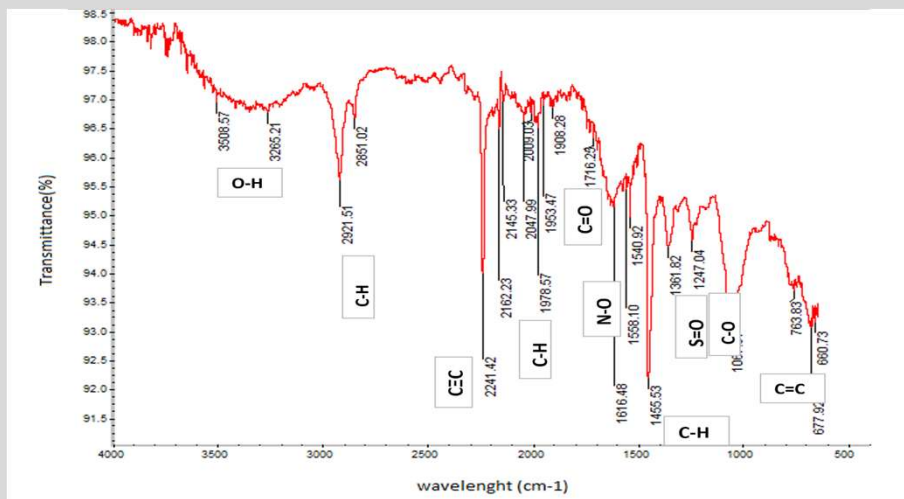


Fig : 19 Chemically Modified PAN fiber After Adsorption

Results and Discussion

➤ Fourier Transform Infrared Spectroscopy



Chemically Modified PAN fiber After Adsorption

Frequency Rang	Bond Nature	Appearance of peak
3000-4000	N-H & O-H Stretching Group	Strong
2900	N-H Stretching Amine Group	Strong
2190-2260	C≡C stretching	Weak
2850-2875	C-H Alkane Bond	Medium Strong
1500-2000	Amines, Nitriles, Carboxylic Bonds	Medium Strong
1780-1650	Carbon-Oxygen Double Bond	Strong
1020-1250	C-N stretching	Strong
2260-2220	Carbon-Nitrogen Triple Bond	Strong
650-1000	C=C bending	Strong

Results and Discussion



➤ Scanning Electron Microscopy (SEM)

SEM of Un Modified PAN Fiber as shown in Figure 20 (a , b)

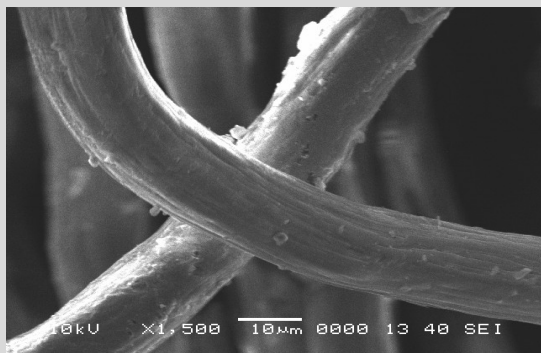


Fig: 20. (a)

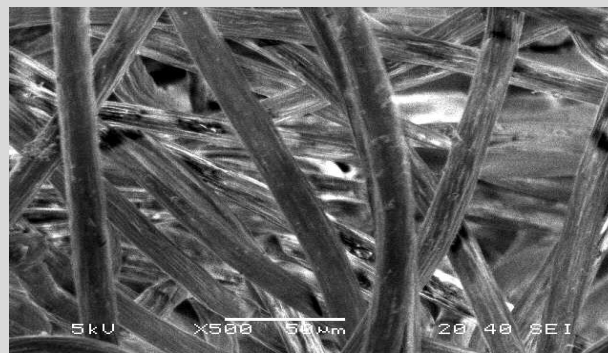


Fig: 20. (b)

Results and Discussion



➤ Scanning Electron Microscopy (SEM)

Chemically Modified PAN fiber After Adsorption as shown in Figure: 21 (a , b)

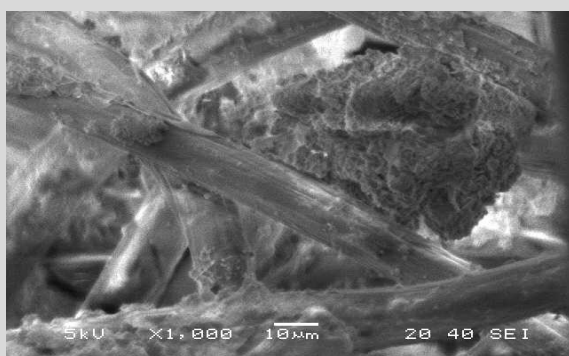


Fig: 21. a

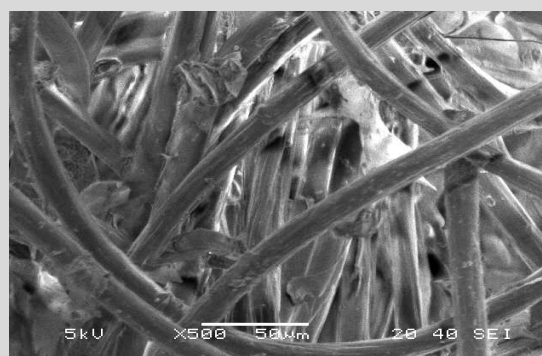


Fig: 21. b

Results and Discussion



As(V) removal against different bed heights: Synthetic arsenate water before Modification of PAN Fiber

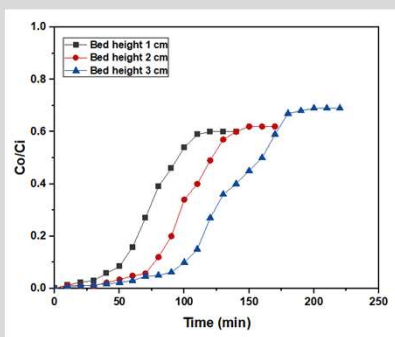


Fig : 22 (a) Feed flowrate 5 ml/min

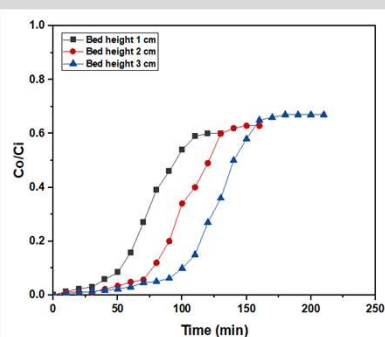


Fig : 22 (b) Feed flowrate 10 ml/min

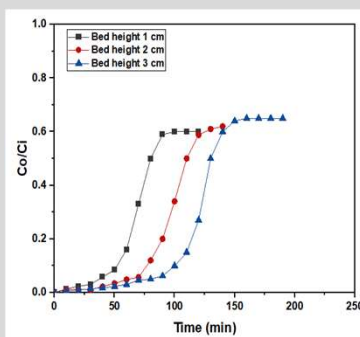


Fig : 22 (c) Feed flowrate 15 ml/min

Fig:22. (a,b,c) In this figure, effect of absorbent bed height (1cm, 2cm, and 3cm) and sampling time from 0 to 250 mints arsenic removal efficiency were examined at flow of 5ml/min, 10ml/min and 15ml/min feed flow rate.

Results and Discussion



As(V) removal against different Flowrates: Synthetic arsenate water before Modification of PAN Fiber

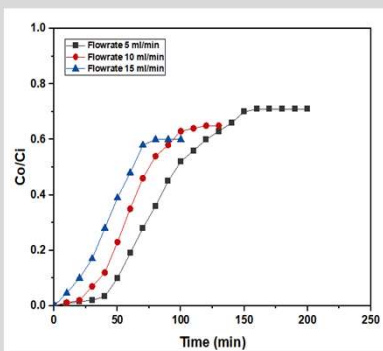


Fig : 23 (a) Bed height 1 cm

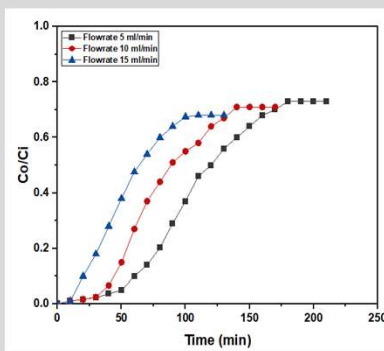


Fig : 23. (b) Bed height 2 cm

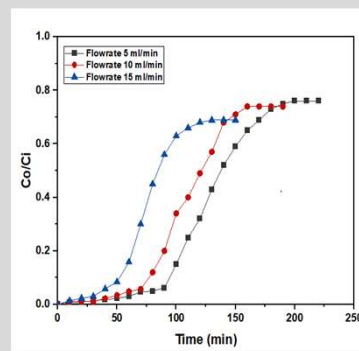


Fig : 23. (c) Bed height 3 cm

Fig:23.(a,b,c) In this figure, effect of absorbent at different flow rates (5ml/min, 10ml/min and 15ml/min) and sampling time from 0 to 250 mints arsenic removal efficiency were examined at bed height 1cm, 2cm and 3cm.

Results and Discussion



As(V) removal against different bed heights: Synthetic arsenate water after Chemical Modification of PAN Fiber

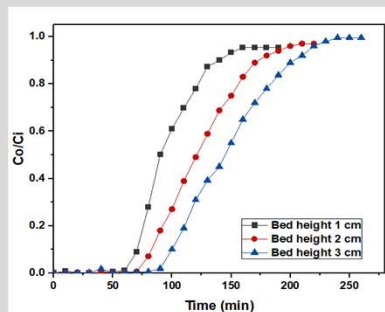


Fig : 24. (a) Feed flowrate 5 ml/min

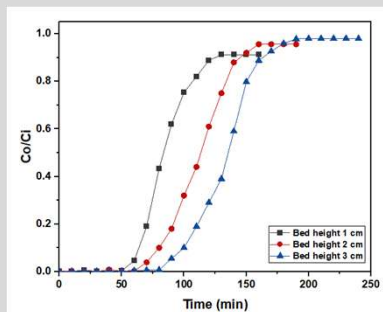


Fig : 24. (b) Feed flowrate 10 ml/min

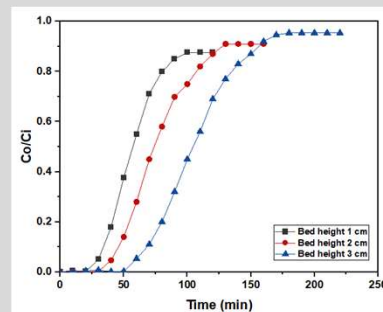


Fig : 24. (c) Feed flowrate 15 ml/min

Fig:24. (a,b,c) In this figure, effect of absorbent bed height (1cm, 2cm, and 3cm) and sampling time from 0 to 250 mints arsenic removal efficiency were examined at flow of 5ml/min, 10ml/min and 15ml/min feed flow rate.

Results and Discussion



As(V) removal against different Flowrates: Synthetic arsenate water after Chemical Modification of PAN Fiber

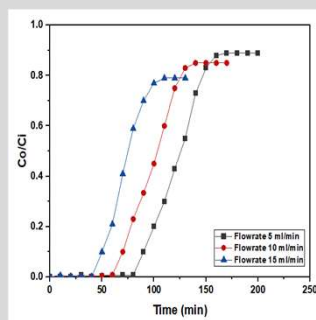


Fig:25. (a) Bed height 1 cm

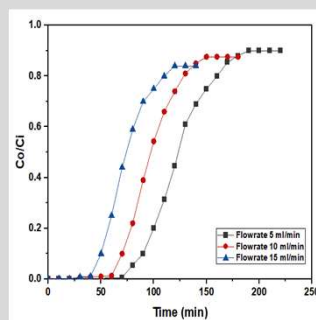


Fig: 25. (b) Bed height 2 cm

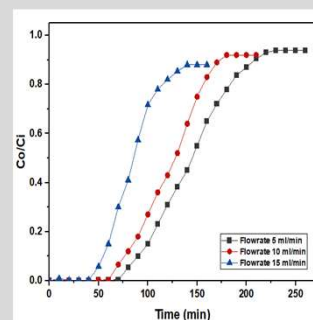


Fig: 25. (c) Bed height 3 cm

Fig:19.(a, b & c) In this figure, effect of absorbent at different flow rates (5ml/min, 10ml/min and 15ml/min) and sampling time from 0 to 250 mints arsenic removal efficiency were examined at bed height 1cm, 2cm and 3cm.

Results and Discussion

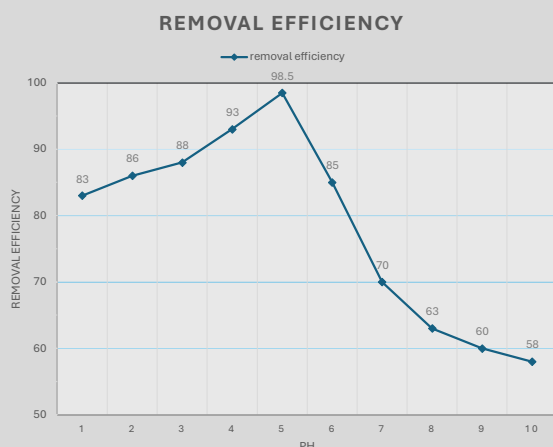


Figure.26. Effect of pH at flowrate 5 ml/min, bed height 3cm and As(V) initial concentration 50 mg/L.

- Arsenate ions (H_3AsO_4) at pH 2, only physical adsorption driving forces between H_3AsO_4 and the adsorbent, resulting in less adsorption. In the range of pH 3 to 7, the predominant species is H_2AsO_4^- .
- The amount of H_2AsO_4^- reaches the maximum at pH 5, which leads to the strongest electrostatic interaction between H_2AsO_4^- and protonated-amino groups on the surface of functionalized PAN fiber so that the removal efficiency reaches 98.5%.
- At higher pH, OH^- ions in the solution can compete with the arsenate ions and so adsorption of As (V) ions reduced.

Results and Discussion



Efficiency Comparison with Before and After Modification of synthetic water Removal

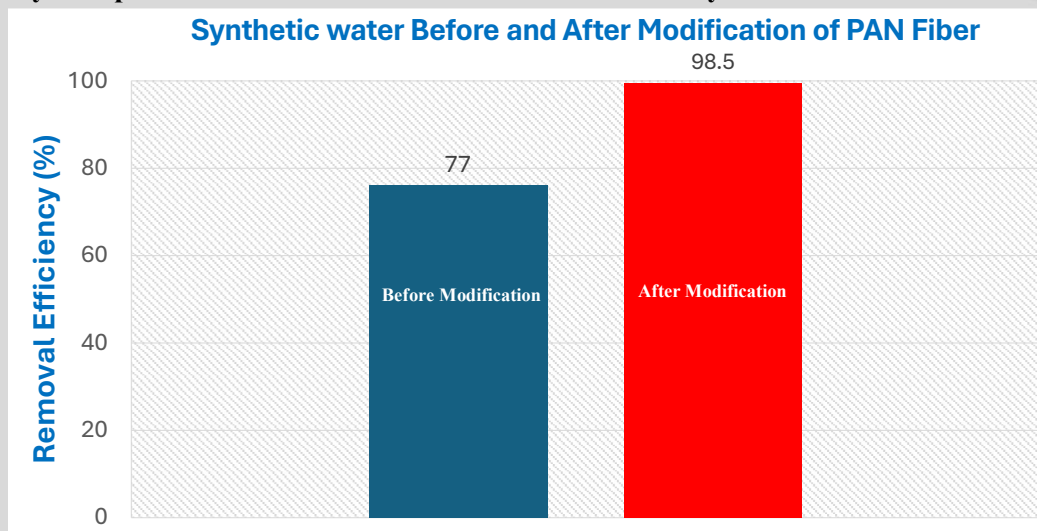


Figure:27. Efficiency comparison b/w before and after Chemical Modifications PAN

Results and Discussion



Efficiency Comparison with pervious research

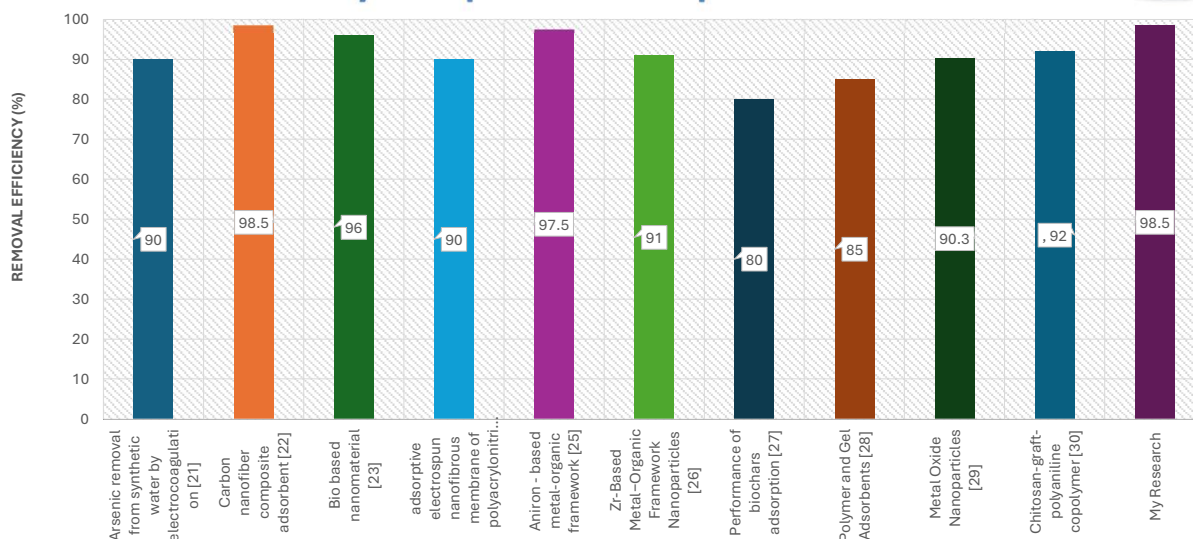


Figure: 28 Graph show the removal efficiency of arsenic from different researchers on different adsorbents

Conclusion



- Modified PAN fiber as an efficient adsorbent was prepared for the removal of toxic Arsenic from synthetic arsenate water to perform lab experiment in a fixed bed column method.
- In this study Modified PAN fiber was characterized by using SEM and FTIR techniques. Arsenic adsorption in the fixed bed column with this adsorbent had breakthrough curve profiles while varying different adsorption parameters, such as influent flow rate, adsorbent bed height, initial solution pH.

Conclusion



- Un Modified PAN Fiber achieved maximum As(V) removal efficiency from synthetic water up to 77% with the optimum adsorbent bed height of 3 cm, pH 5, and 5mL/min at room temperature because of more contact time with adsorbent.
- Chemical Modified PAN Fiber achieved maximum (As) removal efficiency from synthetic water up to 98.5%. Which means Modified PAN Fiber is a sufficient adsorbent for (As) removal.
- The Chemically Modified adsorbent was ideal for household water in rural areas and found to be a cost-effective, bio- degradable, and efficient adsorbent for the removal of arsenic.

Future Recommendations



- The research could be extended in following directions:
- Modified PAN fiber can also be used for the elimination of other toxic metals such as lead, mercury, fluoride etc. to check the removal efficiency of this adsorbent for these metals.
- The adsorption bed could be scaled-up by enhancing the column and bed dimensions.
- Simulation and modelling work to be carried out using the data generated in this study to design arsenic treatment filters.
- Pilot plant of arsenic treatment filter containing iron ore or PAN fiber adsorbent should be installed in the arsenic affected areas of Sindh such as Nawabshah, Sakrand and Khairpur District.

Reference



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